

Mechanistic Interpretability of Vision Transformers: Sparse Autoencoder Feature Decomposition for Visual Concept Discovery

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ABSTRACT

We train Sparse Autoencoders (SAEs) on the MLP output activations of a pre-trained ViT-B/16 at two network depths, and use CLIP as an external cross-modal evaluator to assign interpretable labels to discovered sparse directions—a strategy designed to address the labelling gap that arises in purely visual transformers trained without text supervision. Layer 6 features encode low-level texture and geometric patterns with negligible causal impact (mean relative logit drop: 0.03%); Layer 11 features encompass more semantic concepts and are substantially more causal (mean: 1.40%), concentrated on a single feature (*honeycomb pattern*, 13.49% logit drop under ablation). These results confirm that SAE-based decomposition transfers to purely visual transformers and that external VLM evaluation is a viable labelling strategy.

KEYWORDS

Mechanistic Interpretability, Vision Transformers, Sparse Autoencoders, Polysemanticity, Feature Decomposition, Explainable AI

1 INTRODUCTION

As neural networks are deployed in critical settings, mechanistic interpretability aims to reverse-engineer their computations into analyzable units [6]. Sparse Autoencoders (SAEs) have proven effective at decomposing polysemantic MLP activations of language models into monosemantic features [1, 4]. Applying this to Vision Transformers (ViTs) [5] introduces two specific challenges absent in the language setting: the spatial structure of patch-based attention makes causal tracing harder to formalise, and—crucially—ViTs trained without language supervision provide no built-in mechanism for translating sparse visual directions into human-readable concept labels. This project addresses both challenges by training SAEs on a ViT-B/16 and using CLIP [9] as an external evaluator for cross-modal concept labelling, providing the first systematic evidence that mechanistic interpretability tools transfer to purely visual transformers.

2 RELATED WORK

2.1 Mechanistic Interpretability in Transformers

Elhage et al. [6] formalised the residual stream framework: attention heads route information between tokens while MLP blocks process within-token representations. Building on this, Conmy et al. [3] introduced ACDC, which identifies the minimal computational subgraph responsible for a target behaviour via iterative activation

patching. SAEs applied to MLP activations or residual stream vectors recover a larger dictionary of sparse, monosemantic features that influence model predictions in interpretable ways [1, 4]. These contributions target language models exclusively.

2.2 ViTs and Concept Discovery in Vision-Language Models

Raghu et al. [10] showed that ViTs aggregate global information early via self-attention and propagate features strongly through residual connections—properties that complicate circuit-level analysis relative to language models. In the vision-language domain, Gandelsman et al. [7] decompose CLIP image representations as sparse sums of text-direction embeddings; Haque et al. [8] apply SAEs to medical VLMs and anchor discovered concepts to clinical ontologies. Both approaches rest on pre-existing text-image alignment unavailable in purely supervised ViTs.

3 RESEARCH GAPS

Three gaps motivate this work. **Systematic SAE analysis for pure vision models:** SAEs have not been applied to ViTs trained without language supervision; visual tokens encode image patches rather than words, and there is no direct way to extract a human-understandable label from a learned sparse direction, making it an open empirical question whether SAEs recover meaningful structure from visual activations. **Spatial structure of ViT attention:** unlike language transformers, ViTs operate over all patch pairs simultaneously with no preferred direction of information flow [10]; tracing how evidence from spatial regions is compressed into the [CLS] token requires causal tools not yet validated in the visual setting. **Cross-modal labelling:** when no internal text-image alignment exists, assigning interpretable labels to SAE features requires an external grounding mechanism—a bottleneck unacknowledged in most vision interpretability work.

These gaps converge on one open question: can SAEs recover interpretable, monosemantic features from a purely visual ViT, and how can those features be evaluated without built-in language supervision? This project directly targets that question by using CLIP [9] as a detached external evaluator, decoupling the interpretability analysis from the backbone under study.

4 METHODOLOGY

Our pipeline has four stages.

Backbone and data. We analyse ViT-B/16 [5] (12 transformer blocks, $d=768$, trained on ImageNet-1k) using 1,000 validation images paired with an 18-concept CLIP vocabulary spanning object

Table 1: SAE metrics and mean causal impact. $\bar{L}_0$: avg. active features per patch token.

Layer	R ²	$\bar{L}_0$	Mean RLD
6 (mid)	0.9957	201.72	0.03%
11 (late)	0.9477	107.90	1.40%

categories (*animal, bird, plant, ...*), textures (*fur texture, feather texture, ...*), and geometric patterns (*spotted, honeycomb, scale, ...*).

Activation extraction. Forward hooks collect MLP output activations at Layer 6 (mid-network) and Layer 11 (late-network). Each image yields a 196×768 patch-token matrix per layer ([CLS] excluded), centred by the dataset-level mean before SAE training.

Sparse Autoencoder. Given activation $\mathbf{x} \in \mathbb{R}^d$, the SAE produces a sparse code

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}), \quad \mathbf{z} \in \mathbb{R}^m, \quad (1)$$

and reconstructs $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z}$ (unit-norm columns, $m=8d=6144$). Training minimises

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1 \quad (2)$$

(Adam, $\text{lr}=10^{-3}$, $\lambda=10^{-3}$, batch 256; 5 epochs for Layer 6, 15 for Layer 11).

CLIP labelling. The $K=5$ highest-activating crops per feature are matched to the vocabulary via mean cosine similarity to “a photo of {concept}” prompts [9]. A feature is characterised as monosemantic when the CLIP-assigned label is confirmed by visual inspection of the top-activating crops.

Causal verification. Feature k is ablated by subtracting its contribution from patch-token activations: $\mathbf{x}' = \mathbf{x} - f_k(\mathbf{x}) \mathbf{W}_{\text{dec}}[:, k]$. Causal impact is quantified via the Relative Logit Drop (RLD):

$$\text{RLD} = \frac{L_{\text{baseline}} - L_{\text{ablated}}}{|L_{\text{baseline}}|}. \quad (3)$$

A dose-response sweep over ablation strength $s \in [0, 1]$ rules out artefacts of any single intervention point.

5 EXPERIMENTS AND ANALYSIS

Table 1 reports SAE quality and mean causal impact per layer.

Layer 6 achieves near-perfect reconstruction (3.3% activation density); its mean logit drop of 0.03% confirms that mid-network features are individually causally negligible. Layer 11 has sparser activations (1.8% density) and a mean logit drop 44× higher, concentrated on a single feature. Table 2 shows the top-5 features per layer by CLIP confidence.

5.1 Feature Variation Across Depths

Layer 6 features are dominated by low-level texture and geometric patterns, consistent with the role of mid-network MLP layers as local structure detectors [10]. At Layer 11, the vocabulary shifts towards semantically abstract concepts (*animal, plant, bird*) alongside texture descriptors. Figure 1 shows the top-5 Layer 11 features by CLIP confidence.

Table 2: Top-5 features per layer. CLIP score: mean cosine similarity to top- K crops. RLD: relative logit drop under full ablation. Steer: logit change under 5× amplification.

Layer	Feature / concept	Score	RLD	Steer
6	4770 spotted pattern	0.267	+0.04%	+0.16%
6	4271 honeycomb pattern	0.265	−0.04%	−0.03%
6	4506 scale pattern	0.254	+0.69%	+1.78%
6	2719 spotted pattern	0.251	+0.56%	+0.80%
6	2045 pineapple	0.249	+0.03%	+0.11%
11	5065 honeycomb pattern	0.297	+13.49%	+5.68%
11	3932 fur texture	0.279	+0.00%	+0.00%
11	965 plant	0.268	+0.00%	+0.00%
11	540 striped pattern	0.258	+0.00%	+0.02%
11	1767 animal	0.257	+0.02%	+0.03%

5.2 Causal Analysis

Feature 5065 at Layer 11 (*honeycomb pattern*) accounts for 13.49% relative logit drop under ablation and 5.68% increase under amplification; all remaining nine Layer 11 features show drops below 0.4%. Figure 2 confirms the spatial selectivity of the feature and the genuineness of its causal contribution via a monotonically increasing dose-response curve.

5.3 Discussion

The results address both research gaps. SAEs recover structured, labellable features from purely visual MLP activations: all features at both depths receive distinct CLIP labels, confirmed by visual inspection. Human assessment independently validated CLIP labels for texture-dominant features; convergence on Feature 5065 (*honeycomb pattern*)—from both CLIP scoring and independent human review, combined with the 13.49% logit drop—constitutes the strongest evidence of monosemantic, causally relevant behaviour. The depth-wise shift from texture detectors to semantic features reproduces the representational hierarchy expected from prior ViT analyses [10]. The concentration of causal impact on a single late-network feature is the most unexpected finding; it may reflect specialisation in the [CLS] accumulation mechanism, though a larger image sample is needed to confirm whether this pattern is input-specific or systematic.

6 CONCLUSIONS

SAE-based feature decomposition transfers to purely visual Vision Transformers: SAEs trained on ViT-B/16 MLP activations recover sparse, interpretable features at two depths, and CLIP-based external evaluation provides a viable labelling strategy without internal text supervision. Mid-network features (Layer 6) are dominated by texture detectors with negligible causal footprint; late-network features (Layer 11) encompass more semantic concepts, with causal impact concentrated on a single highly specialised feature (Feature 5065, 13.49% logit drop). Limitations include the finite and ImageNet-biased CLIP vocabulary, the restriction to MLP outputs (attention heads and residual stream remain unexplored), and the qualitative nature of the monosemanticity assessment. Natural extensions include circuit-level activation patching to trace

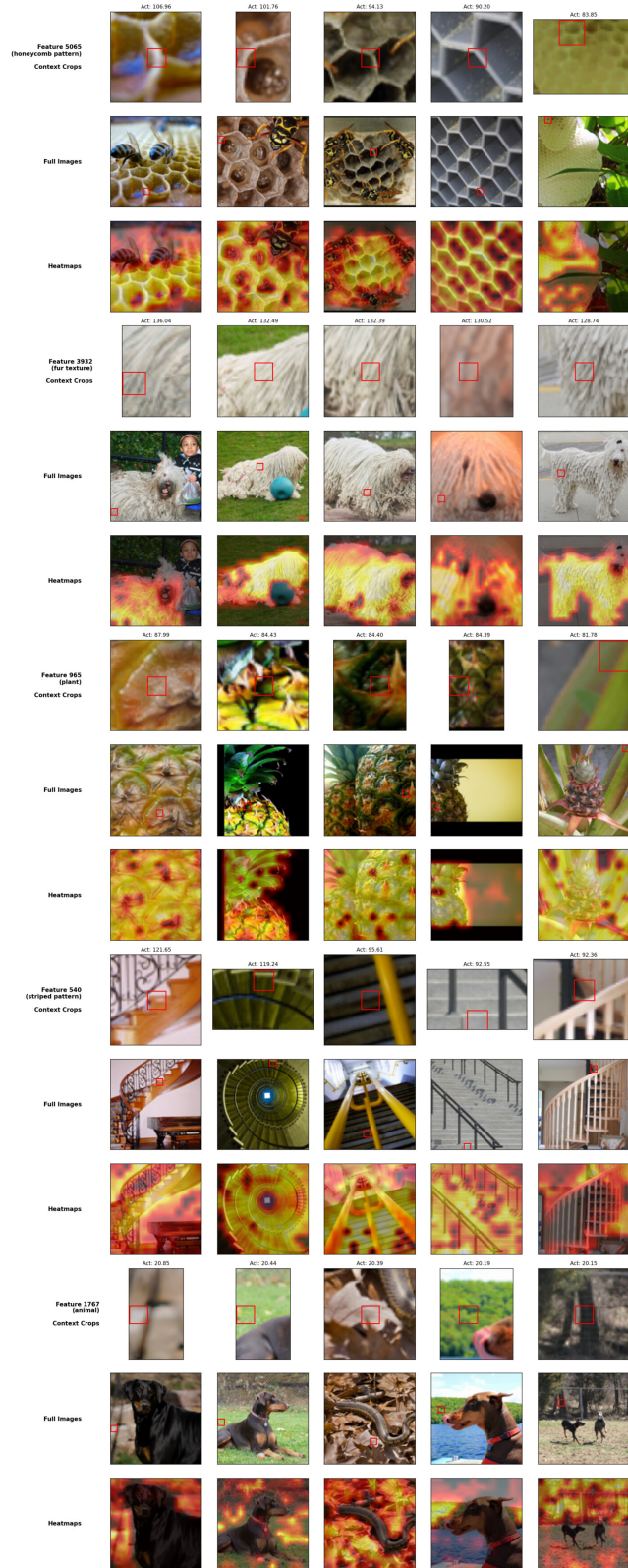


Figure 1: Top-5 Layer 11 features. Each row: contextual crops, full images with highlighted patch, activation heatmaps. Feature 5065 (*honeycomb pattern*) shows the highest causal impact (13.49% RLD).

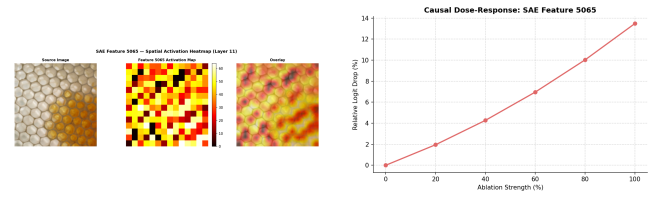


Figure 2: Feature 5065. Left: spatial activation heatmap showing selective activation on coherent regions. Right: dose-response curve—monotonically increasing logit drop as ablation strength goes from 0 to 1.

classification-specific computation, and a comparative analysis of SAE dictionaries from supervised versus DINO-pretrained ViTs [2]—two regimes known to produce qualitatively different internal representations [10].

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